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(54) **THREE-DIMENSIONAL MEMORY DEVICES
AND FABRICATING METHODS THEREOF**

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None

See application file for complete search history.

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Primary Examiner — Sue A Purvis

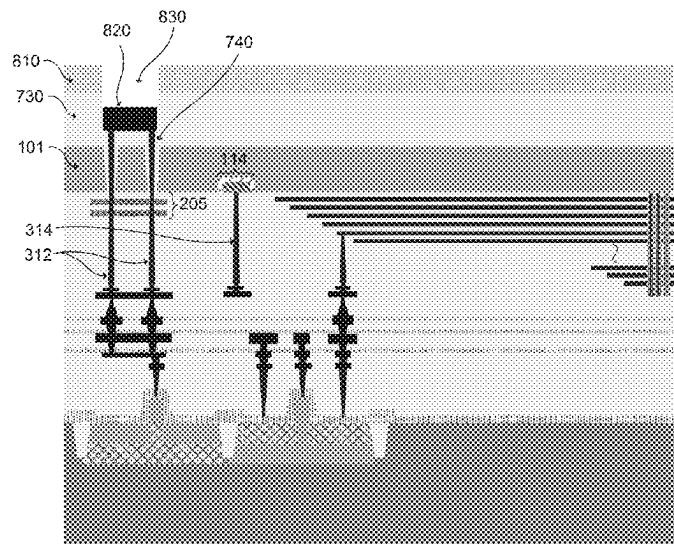
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(57) **ABSTRACT**

A method for forming a gate structure of a 3D memory device is provided. The method comprises forming an etch stop structure in a first wafer, forming a first through contact in contact with the etch stop structure, bonding the first wafer to a second wafer to electrically connect the first through contact to a CMOS device of the second wafer, and forming a through substrate contact penetrating a first substrate of the first wafer and the etch stop structure, and in electrically contact with the CMOS device through the first through contact.

20 Claims, 13 Drawing Sheets



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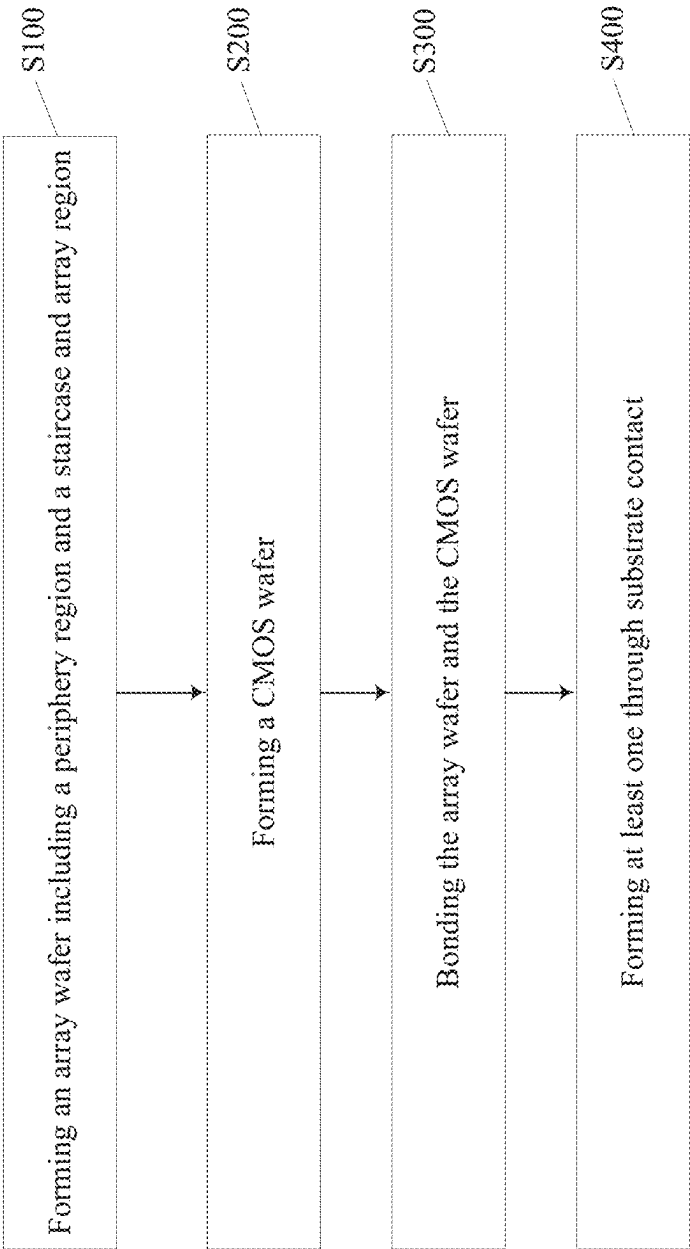


FIG. 1A



FIG. 1B

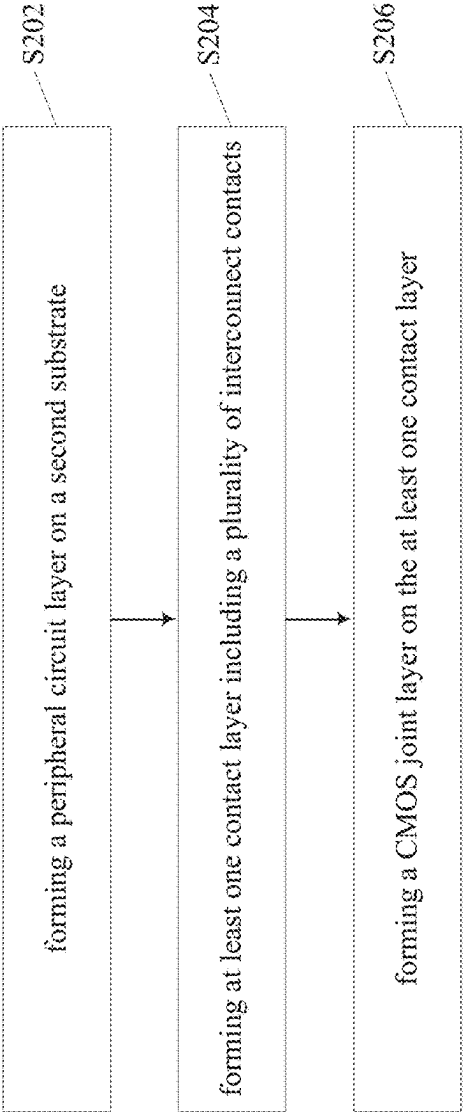


FIG. 1C

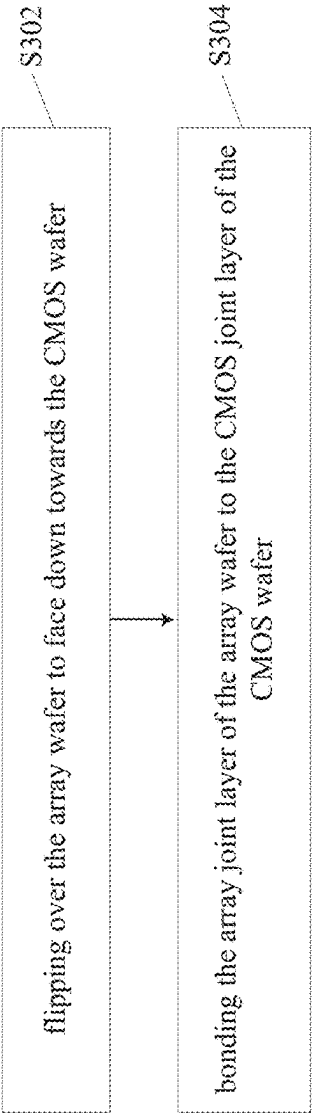


FIG. 1D

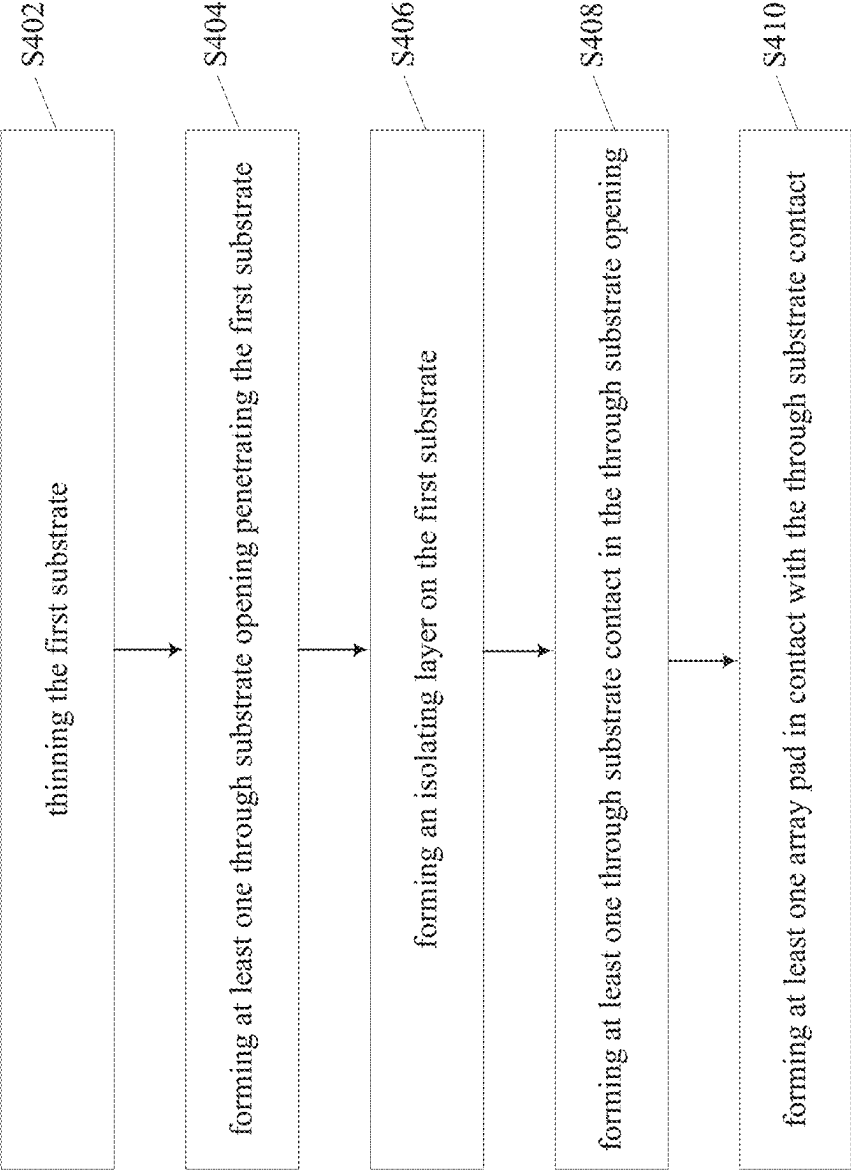


FIG. 1E

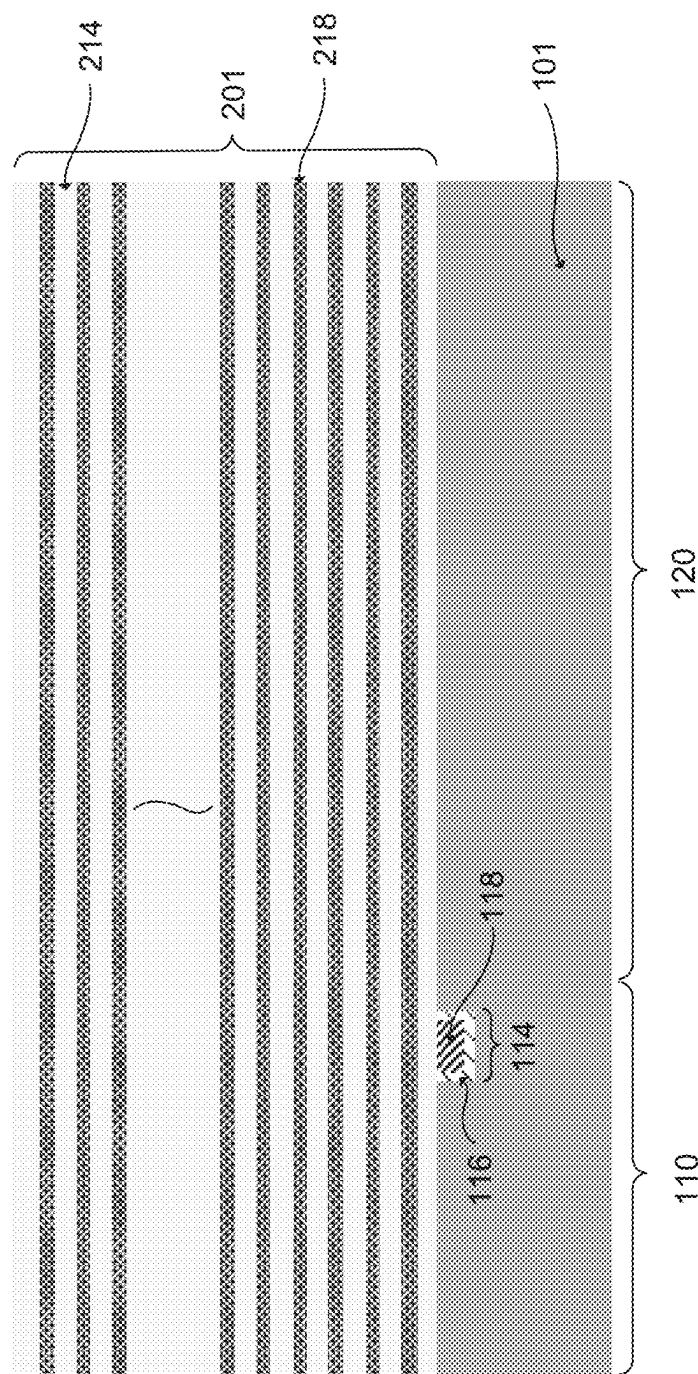
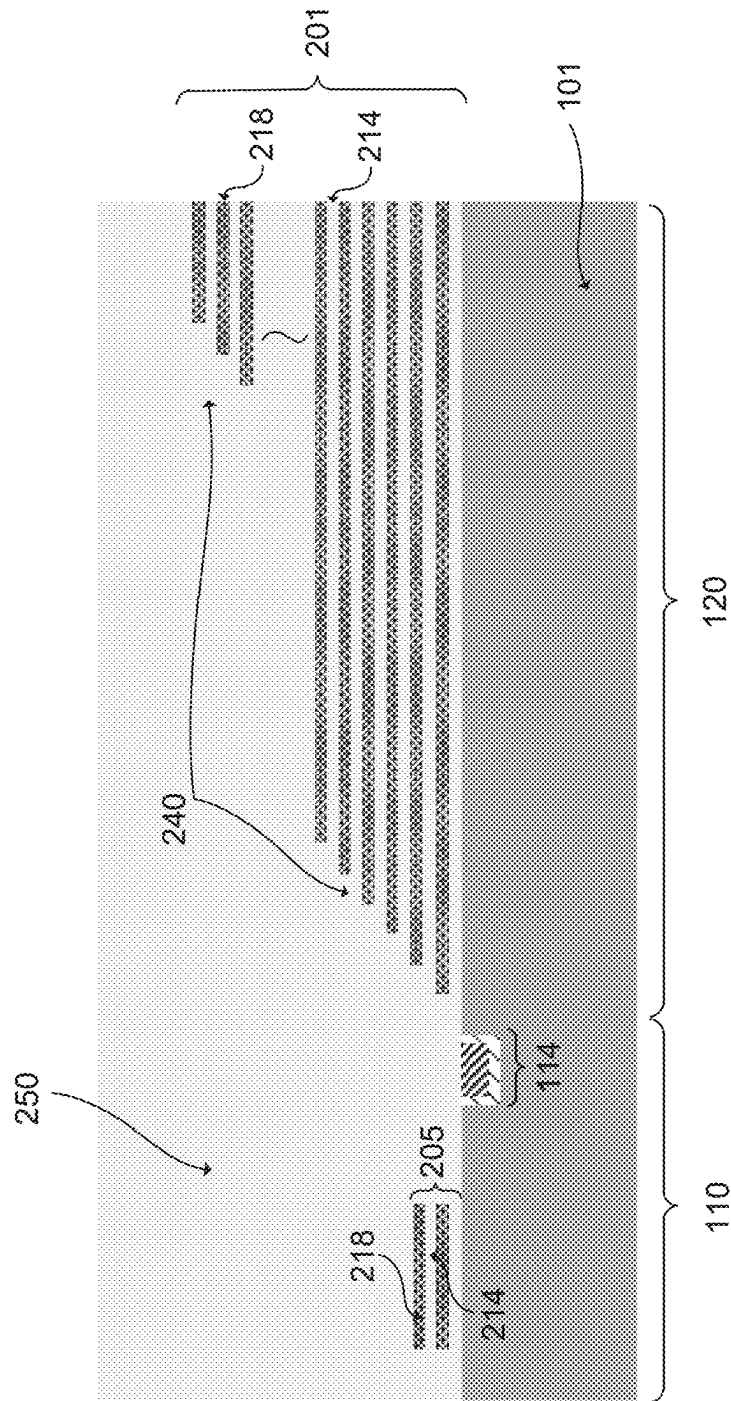


FIG. 2



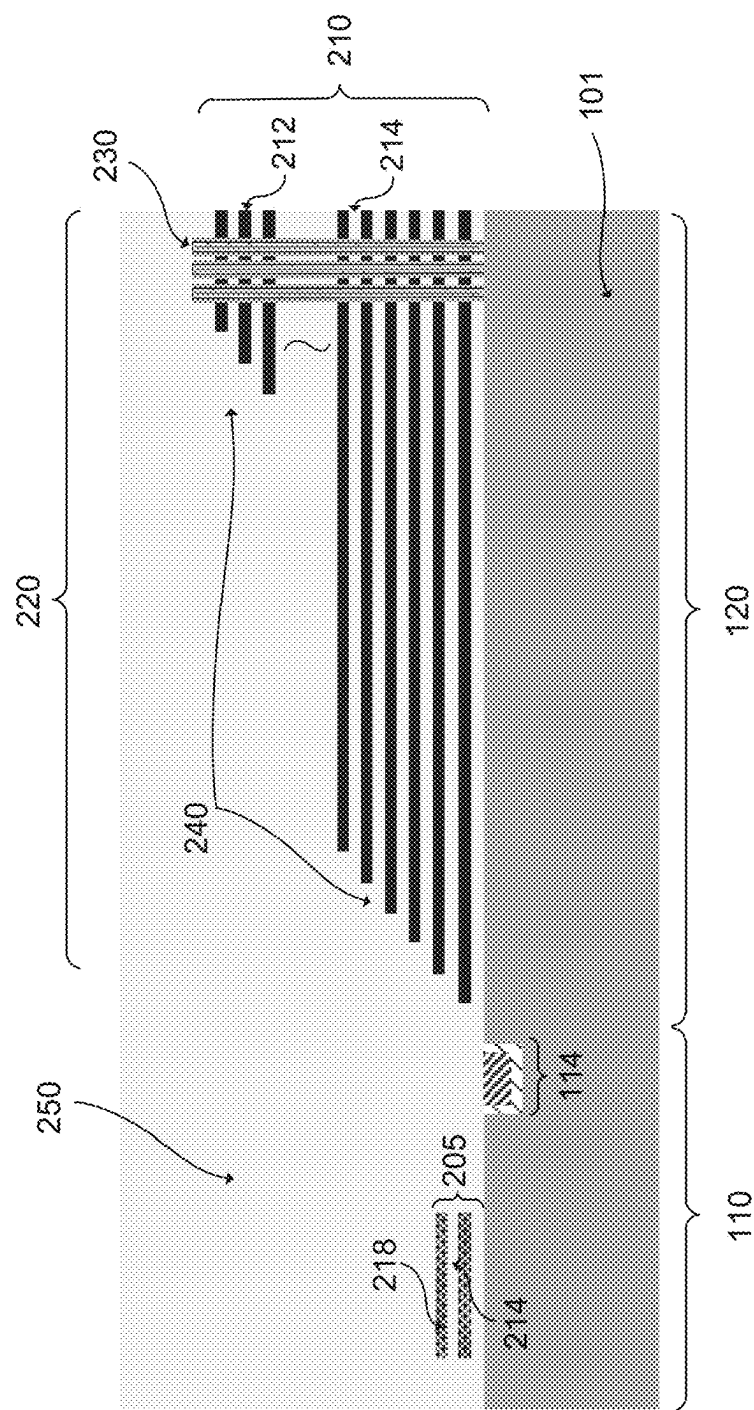


FIG. 4

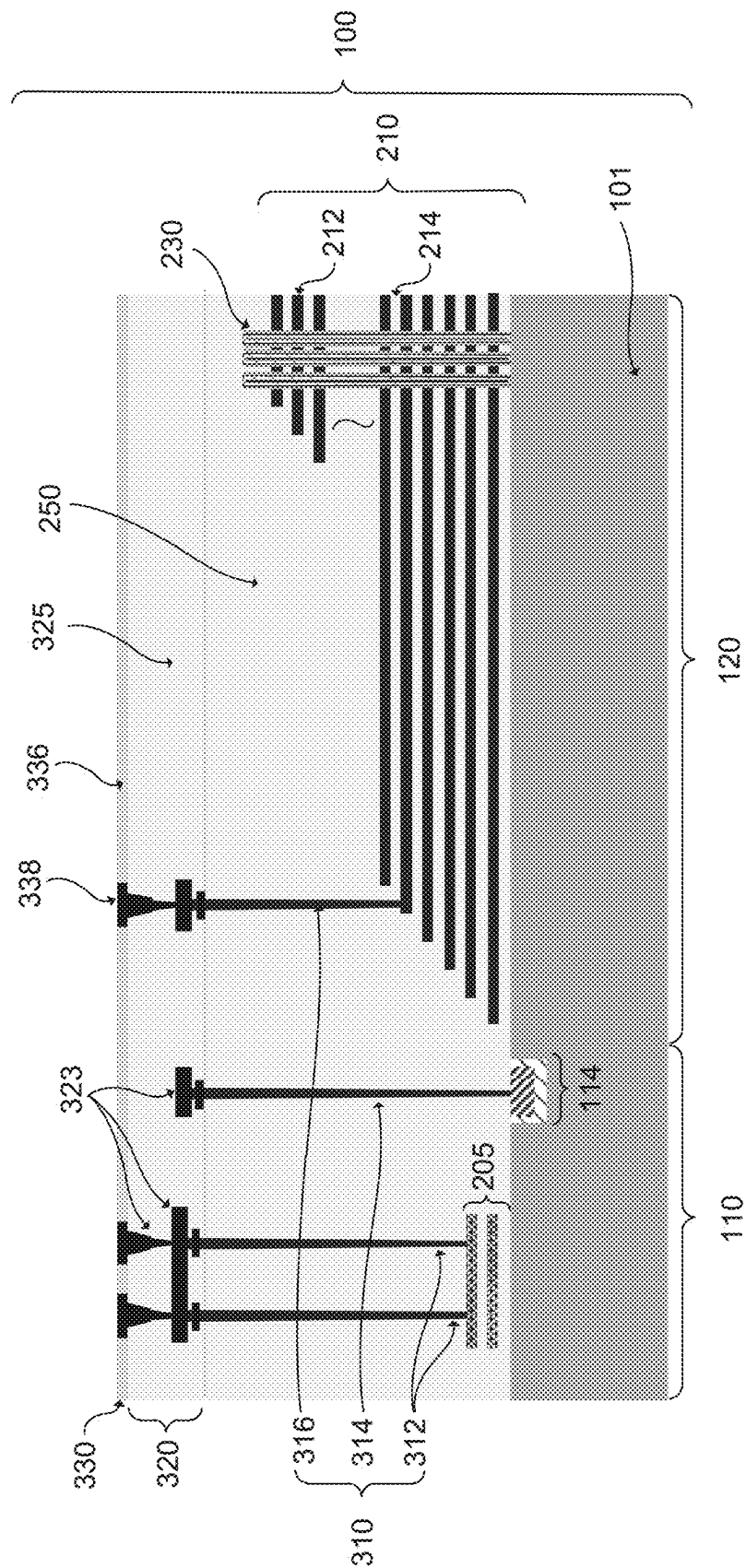


FIG. 5

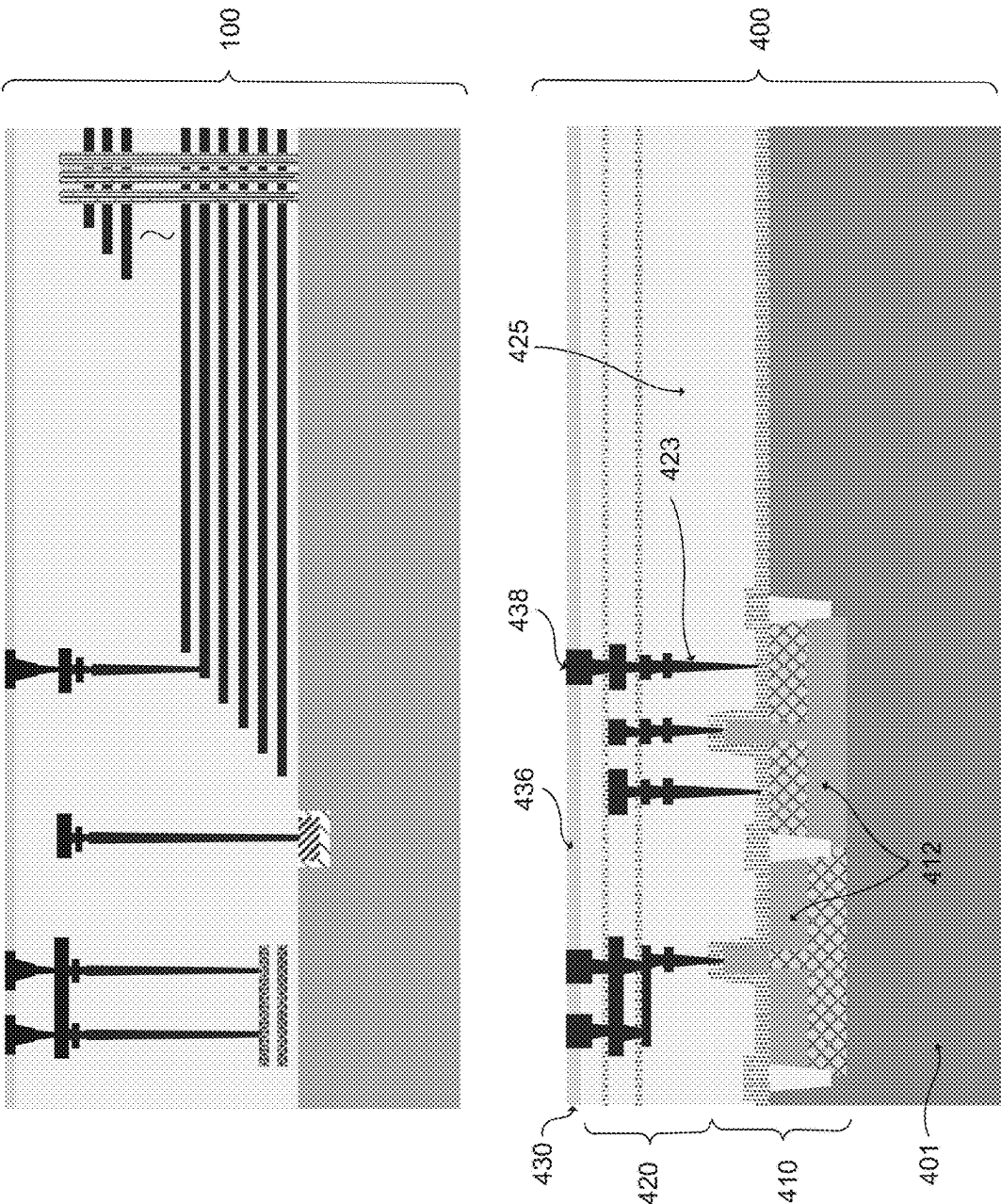


FIG. 6

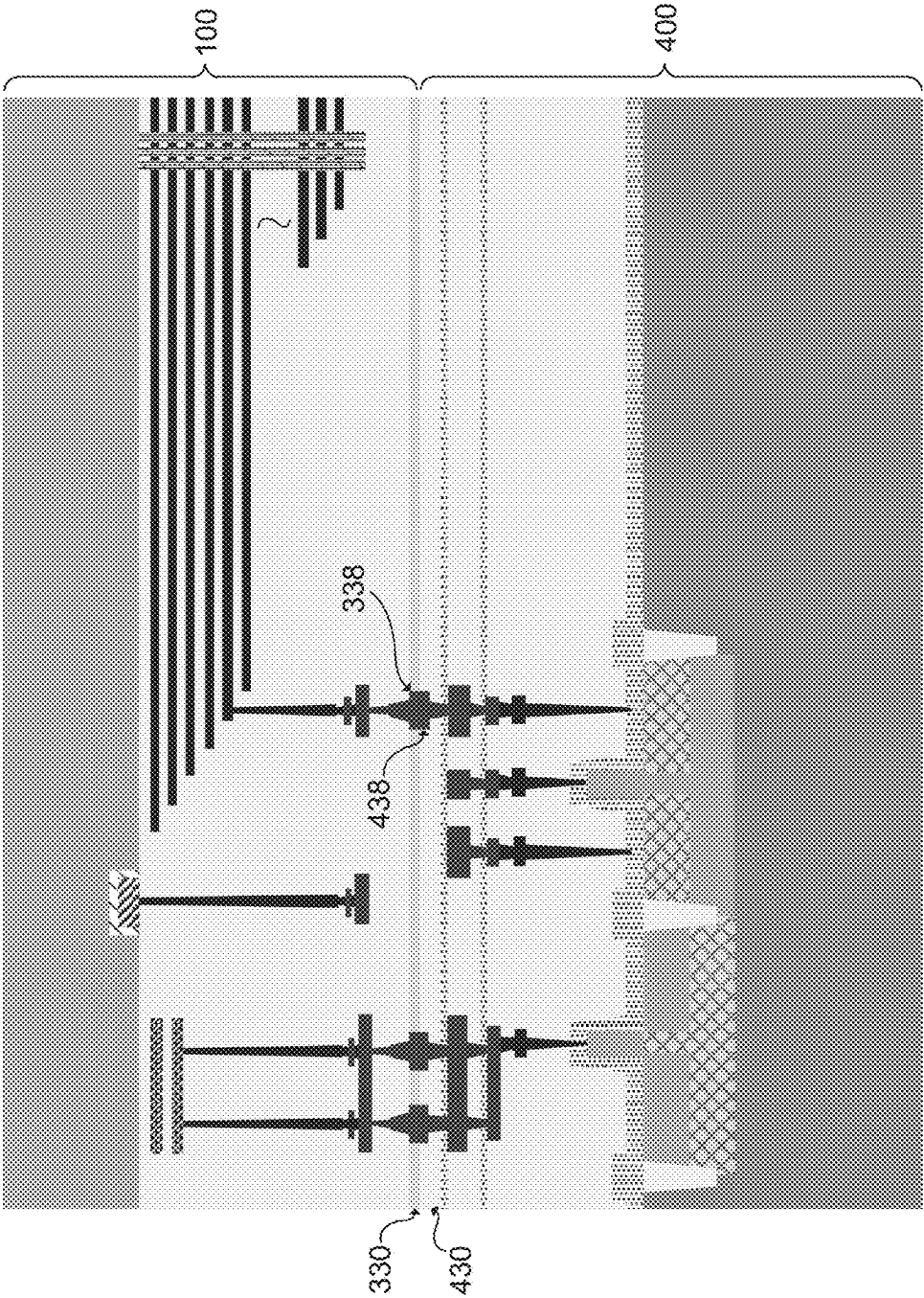


FIG. 7

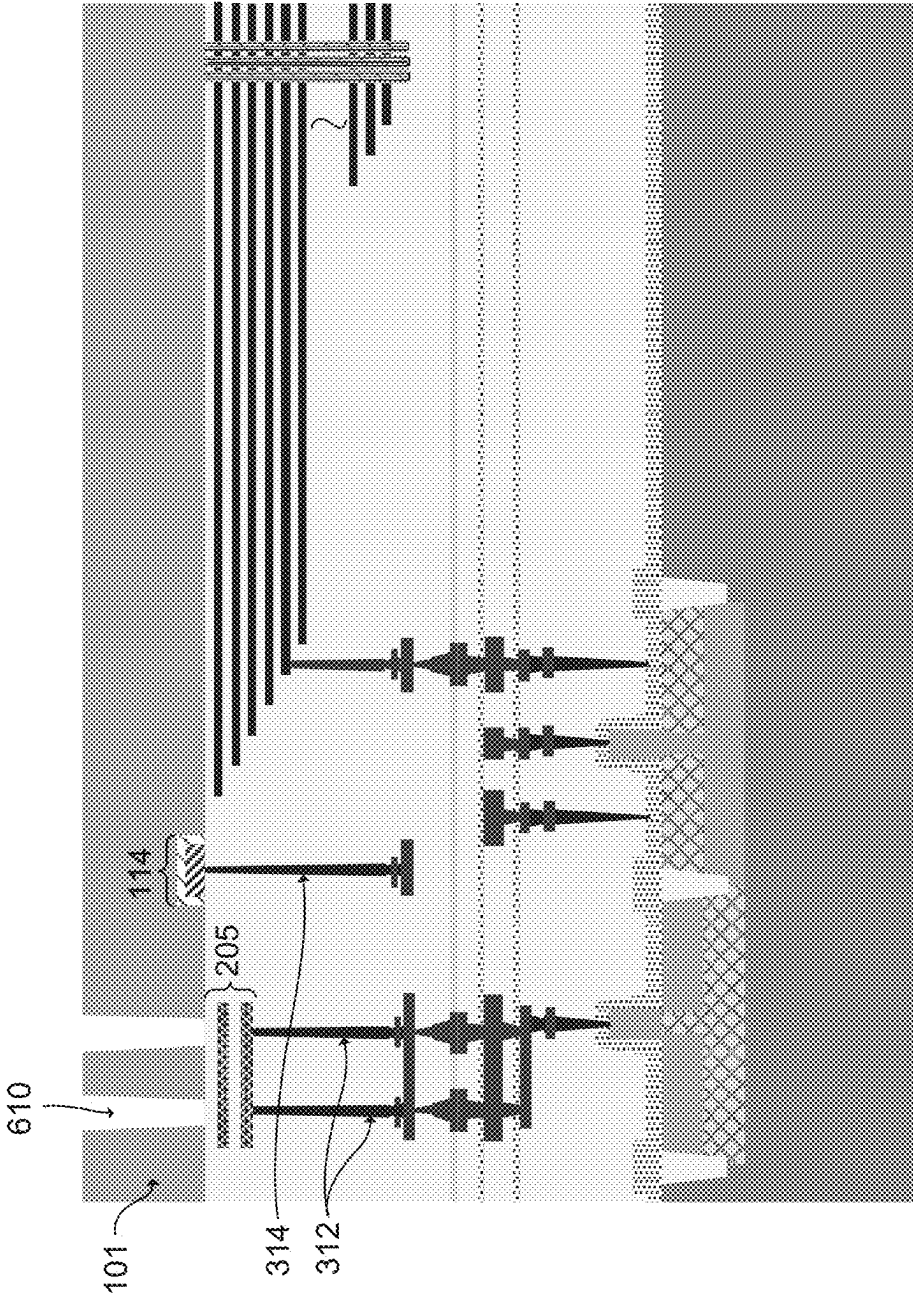


FIG. 8

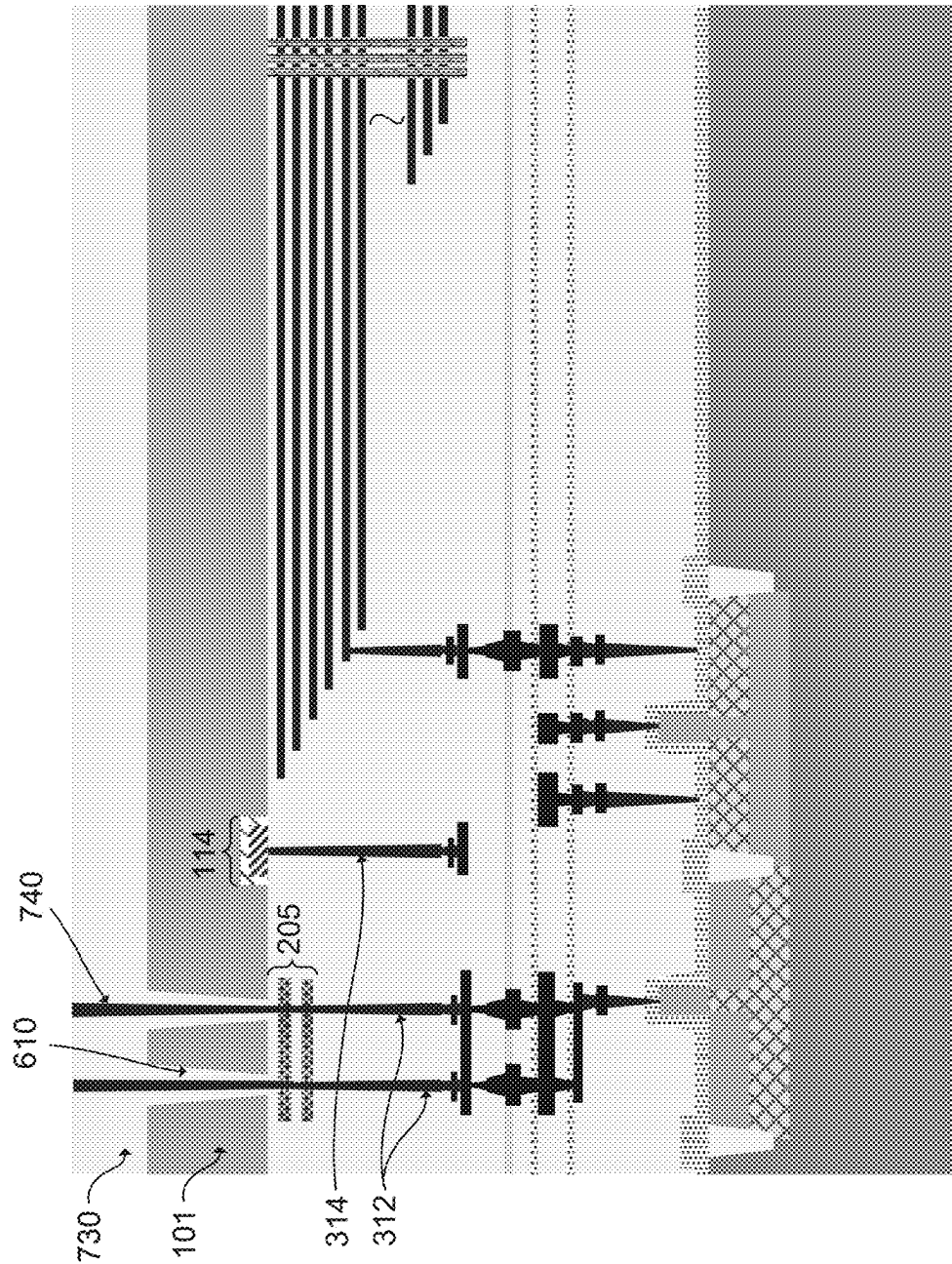
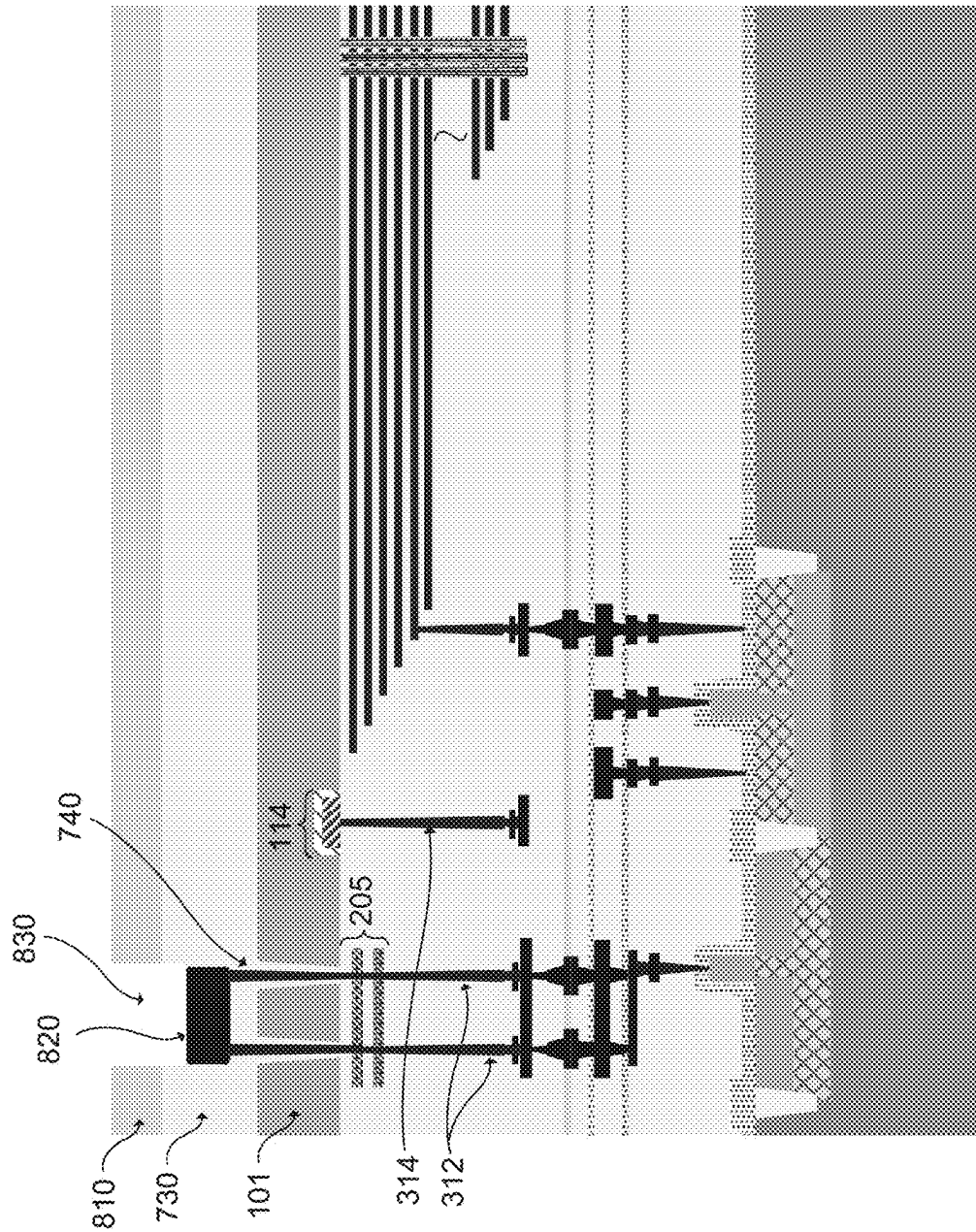


FIG. 9



OFFICE

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THREE-DIMENSIONAL MEMORY DEVICES AND FABRICATING METHODS THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. patent application Ser. No. 16/729,865, filed on Dec. 30, 2019, which claims priority to PCT Application No. PCT/CN2019/099031, filed on Aug. 2, 2019, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

Embodiments of the present disclosure relate to three-dimensional 3D memory devices and fabrication methods thereof.

BACKGROUND

Planar memory cells are scaled to smaller sizes by improving process technology, circuit designs, programming algorithms, and fabrication processes. However, as feature sizes of the memory cells approach a lower limit, planar processes and fabrication techniques become challenging and costly. As a result, memory density for planar memory cells approaches an upper limit.

A 3D memory architecture can address the upper density limitation in planar memory cells. The 3D memory architecture includes a memory array and peripheral devices for controlling signals to and from the memory array.

BRIEF SUMMARY

Embodiments of method for forming gate structures of 3D memory devices and fabrication methods thereof are disclosed herein.

Disclosed is a method for forming a three-dimensional (3D) NAND memory device, comprising forming an array wafer including a periphery region and a staircase and array region, comprising: forming an alternating dielectric etch stop structure on a first substrate in the periphery region, forming an array device on the first substrate in the staircase and array region, and forming at least one first vertical through contact in the periphery region and in contact with the alternating dielectric etch stop structure. The method can further include: forming a CMOS wafer; bonding the array wafer and the CMOS wafer; and forming at least one through substrate contact penetrating the first substrate and the alternating dielectric etch stop structure, and in contact with the at least one first vertical through contact.

In some embodiments, forming the array wafer further comprises: forming an array well structure in the first substrate in the periphery region; and forming at least one second vertical through contact in contact with the array well structure.

In some embodiments, forming the array wafer further comprises: forming an alternating dielectric stack on the first substrate; and removing a portion of the alternating dielectric stack to simultaneously form the alternating dielectric etch stop structure in the periphery region and a staircase structure on at least one lateral side of the alternating dielectric stack in the staircase and array region.

In some embodiments, forming the array device comprises: converting the alternating dielectric stack in the staircase and array region to an alternating conductor/di-

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electric stack; and forming a plurality of NAND strings vertically penetrating the alternating conductor/dielectric stack.

In some embodiments, forming the array wafer further comprises: forming an insulating layer covering the alternating dielectric etch stop structure, the array well structure, and the array device; and forming at least one word line contact in the staircase and array region and in contact with a word line in the staircase structure; wherein the at least one first vertical through contact, the at least one second vertical through contact, and the at least one word line contact are simultaneously formed in the insulating layer by a same contact forming process.

In some embodiments, forming the array wafer further comprises: forming at least one first contact layer including a plurality of first interconnect contacts on the insulating layer; and forming an array joint layer on the at least one first contact layer.

In some embodiments, forming the CMOS wafer comprises: forming a peripheral circuit layer on a second substrate; forming at least one second contact layer including a plurality of second interconnect contacts on the peripheral circuit layer; and forming a CMOS joint layer on the at least one second contact layer.

In some embodiments, bonding the array wafer to the CMOS wafer comprises: flipping over the array wafer to face down towards the CMOS wafer; and bonding the array joint layer of the array wafer to the CMOS joint layer of the CMOS wafer, such that the at least one first vertical through contact is electrically connected to the peripheral circuit layer through at least one first interconnect contact and at least one second interconnect contact.

In some embodiments, forming the at least one through substrate contact comprises: forming at least one through substrate opening penetrating the first substrate; forming an isolating layer covering the first substrate and filling the at least one through substrate opening; forming at least one vertical through opening that penetrates the isolating layer, the at least one through substrate opening, and the alternating dielectric etch stop structure, and exposes at least a portion of the at least one first vertical through contact; and forming the at least one through substrate contact in the at least one vertical through opening, such that the at least one through substrate contact is in contact with the at least one first vertical contact.

In some embodiments, forming at least one through substrate opening comprises: using a deep plasma etching to form the at least one through substrate opening in the first substrate; wherein a high-energy stream of plasma during the deep plasma etching is blocked by the alternating dielectric etch stop structure and the array well structure.

In some embodiments, forming the at least one through substrate contact further comprises: forming at least one array pad in contact with the least one through substrate contact; and forming at least one pad opening to expose the at least one array pad.

The disclosed three-dimensional (3D) memory device comprises an array wafer including a periphery region and a staircase and array region, comprising: a first substrate, an alternating dielectric etch stop structure on the first substrate in the periphery region, an array device on the first substrate in the staircase and array region, at least one first vertical through contact in the periphery region, and at least one through substrate contact penetrating the first substrate and the alternating dielectric etch stop structure, and in contact with the at least one first vertical through contact. The 3D memory device further comprises a CMOS wafer bonded on

the array wafer, comprising a peripheral circuit layer electrically connected to the at least one through substrate contact through the at least one first vertical through contact.

In some embodiments, the array wafer further comprises: an array well structure in the first substrate in the periphery region; and at least one second vertical through contact in contact with the array well structure.

In some embodiments, the alternating dielectric etch stop structure comprises at least two dielectric layer pairs each including a first dielectric layer and a second dielectric layer different from the first dielectric layer.

In some embodiments, the array device comprises: an alternating conductor/dielectric stack on the first substrate; a plurality of NAND strings vertically penetrating the alternating conductor/dielectric stack; and a staircase structure on at least one lateral side of the alternating conductor/dielectric stack.

In some embodiments, the array wafer further comprises: an insulating layer covering the alternating dielectric etch stop structure, the array well structure, and the array device; at least one word line contact in the staircase and array region and in contact with a word line in the staircase structure; wherein the at least one first vertical through contact, the at least one second vertical through contact, and the at least one word line contact penetrate the insulating layer.

In some embodiments, the array wafer further comprises: at least one first contact layer including a plurality of first interconnect contacts covering the insulating layer; and an array joint layer between the at least one first contact layer and the CMOS wafer.

In some embodiments, the CMOS wafer comprises: a peripheral circuit layer on a second substrate; at least one second contact layer including a plurality of second interconnect contacts on the peripheral circuit layer; and a CMOS joint layer between the at least one second contact layer and the array joint layer.

In some embodiments, the array wafer further comprises: an isolating layer covering the first substrate; wherein the at least one through substrate contact penetrates the isolating layer and the alternating dielectric etch stop structure, and in contact with the at least one first vertical contact.

In some embodiments, the array wafer further comprises: at least one array pad in contact with the least one through substrate contact; wherein the at least one array pad is electrically connected to the peripheral circuit layer of the CMOS wafer through the at least one first vertical through contact, the at least one first interconnect contact, and the at least one second interconnect contact.

Other aspects of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate embodiments of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

FIGS. 1A-1E illustrate flow diagrams of an exemplary method for forming a 3D memory device, according to some embodiments of the present disclosure; and

FIGS. 2-10 illustrate schematic cross-sectional views of an exemplary 3D memory device at certain fabricating

stages of the method shown in FIG. 1 according to some embodiments of the present disclosure.

Embodiments of the present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. A person skilled in the pertinent art will recognize that other configurations and arrangements can be used without departing from the spirit and scope of the present disclosure. It will be apparent to a person skilled in the pertinent art that the present disclosure can also be employed in a variety of other applications.

It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “some embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of a person skilled in the pertinent art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context.

It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

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As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure, or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnection layer can include one or more conductor and contact layers (in which contacts, interconnect lines, and/or vias are formed) and one or more dielectric layers.

As used herein, the term “nominal/nominally” refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the term “about” indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the term “about” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., $\pm 10\%$, $\pm 20\%$, or $\pm 30\%$ of the value).

As used herein, the term “3D memory device” refers to a semiconductor device with vertically-oriented strings of memory cell transistors (i.e., region herein as “memory strings,” such as NAND strings) on a laterally-oriented substrate so that the memory strings extend in the vertical direction with respect to the substrate. As used herein, the term “vertical/vertically” means nominally perpendicular to a lateral surface of a substrate.

As semiconductor technology advances, three-dimensional (3D) memory devices, such as 3D NAND memory devices, keep scaling more oxide/nitride; (ON) layers. Generally, in some conventional methods for forming a 3D memory device, a complementary metal-oxide-semiconductor wafer (“CMOS wafer” hereinafter) is bonded with a memory cell array wafer (“array wafer” hereinafter) to form a framework of the 3D memory device. To form interconnect structures for providing vertical electrically interconnects between the stacked memory cell array devices and peripheral devices (e.g., for power bus and metal routing), a through silicon contact (TSC) etching process is performed to penetrate the entire silicon layer which has a large thickness due to the increased numbers of ON layers. As the etching hole has a substantial aspect ratio, it requires to a quantity of energy to perform a plasma etching process to form the TSC. Further, since the TSC etching process is performed after bonding the CMOS wafer and the memory cell array wafer, the plasma during the TSC etching process can go through multiple layers and damage the CMOS devices, thus effecting the reliability of the CMOS devices.

Accordingly, a new 3D memory device and a fabricating method thereof are provided to address such issues. It is noted that, the 3D memory device can be a part of a non-monolithic 3D memory device, in which components (e.g., the CMOS devices and the memory cell array device) are formed separately on different wafers and then bonded in a face-to-face manner. In some embodiments, as described

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below in connection with FIGS. 1-9, the array device substrate is flipped and faces down towards the CMOS substrate for hybrid bonding, so that in the bonded non-monolithic 3D memory device, the array wafer is above the CMOS wafer. It is understood that in some other embodiments, the array wafer remains as the substrate of the bonded non-monolithic 3D memory device, and the CMOS wafer is flipped and faces down towards the array wafer for hybrid bonding.

Referring to FIGS. 1A-1E, flow diagrams of an exemplary method for forming a 3D memory device are illustrated in accordance to some embodiments of the present disclosure. It should be understood that the operations and/or steps shown in FIGS. 1A-1E are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. FIGS. 2-10 illustrate schematic cross-sectional views of an exemplary 3D memory device at certain fabricating stages of the method shown in FIGS. 1A-1E according to some embodiments of the present disclosure.

As shown in FIG. 1A, the method starts at operation S100, in which an array wafer including a periphery region and a staircase and array region is formed. An alternating dielectric etch stop structure and an array well structure can be formed in the periphery region, and an array device can be formed in the staircase and array region.

Referring to FIG. 1B and FIG. 2, operation S100 can include step S102 of forming a first substrate 101, and forming an array well structure 114 in the periphery region 110. In some embodiments, the first substrate 101 can include any suitable semiconductor material that can include silicon (e.g., monocrystalline silicon, polycrystalline silicon), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon on insulator (SOI), germanium on insulator (GOI), or any suitable combination thereof. In some embodiments, base substrate 540 is a thinned substrate (e.g., a semiconductor layer), which was thinned by grinding, wet/dry etching, chemical mechanical polishing (CMP), or any combination thereof. In some embodiments, the first substrate 101 can be a single layer substrate or a multi-layer substrate, for example, a monocrystalline single-layer substrate, a polycrystalline silicon (polysilicon) single-layer substrate, a polysilicon and metal multi-layer substrate, etc.

The first substrate 101 can include two lateral surfaces (e.g., a top surface and a bottom surface) extending laterally in the lateral direction. As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of a semiconductor device is determined relative to the substrate of the semiconductor device (e.g., first substrate 101) in the vertical direction when the substrate is positioned in the lowest plane of the semiconductor device in the vertical direction. The same notion for describing spatial relationship is applied throughout the present disclosure.

In accordance with some embodiments, as shown in FIG. 2, the first substrate 101 can include a periphery region 110 and a staircase and array region 120 both extending in the lateral direction. The alternating dielectric etch stop structure 205 and the array well structure 114 can be formed in the periphery region 110 of the first substrate 101.

The array well structure 114 can include an n-well region 116 and p-well region 118, thus form a PN junction. The n-well region 116 and p-well region 118 can be formed by any suitable doping processes. The array well structure 114 can cover a second portion of a top surface of the periphery region 110 of the first substrate 101, and has no overlap with the alternating dielectric etch stop structure 205.

Referring to FIG. 1B and FIG. 2, operation S100 can further include step S104 of forming an alternating dielectric stack 201 on the first substrate 101. As shown in FIG. 2, the alternating dielectric stack 201 can include a plurality of dielectric layer pairs each including a first dielectric layer 214 and a second dielectric layer 218 that is different from the first dielectric layer 214. In some embodiments, the first dielectric layers 214 can be used as insulating layers, and the second dielectric layers 218 can be used as sacrificial layers, which are to be removed in the subsequent processes.

The plurality of first dielectric layers 214 and second dielectric layers 218 are extended in a lateral direction that is parallel to a surface of the first substrate 101. In some embodiments, there are more layers than the dielectric layer pairs made of different materials and with different thicknesses in the alternating dielectric stack 201. The alternating dielectric stack 201 can be formed by one or more thin film deposition processes including, but not limited to, Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD), Atomic Layer Deposition (ALD), or any combination thereof.

In some embodiments, the alternating dielectric stack 201 can include a plurality of oxide/nitride layer pairs. Each dielectric layer pair includes an oxide layer 214 and a nitride layer 218. The plurality of oxide/nitride layer pairs are also referred to herein as an “alternating oxide/nitride stack.” That is, in the alternating dielectric stack 201, multiple oxide layers 214 and multiple nitride layers 218 alternate in a vertical direction. In other words, except a top and a bottom layer of a given alternating oxide/nitride stack, each of the other oxide layers 214 can be sandwiched by two adjacent nitride layers 218, and each of the nitride layers 218 can be sandwiched by two adjacent oxide layers 214.

Oxide layers 214 can each have the same thickness or have different thicknesses. For example, a thickness of each oxide layer can be in a range from about 10 nm to about 150 nm. Similarly, nitride layers 218 can each have the same thickness or have different thicknesses. For example, a thickness of each nitride layer can be in a range from about 10 nm to about 150 nm. In some embodiments, a total thickness of the alternating dielectric stack 600 can be larger than 1000 nm. It is noted that, the thickness ranges are provided for illustration, and should not be construed to limit the scope of the appended claims.

It is noted that, in the present disclosure, the oxide layers 214 and/or nitride layers 218 can include any suitable oxide materials and/or nitride materials. For example, the oxide materials and/or nitride materials can have any suitable elements including, but not limited to, tungsten (W), cobalt (Co), copper (Cu), aluminum (Al), doped silicon, silicides, or any combination thereof. In some embodiments, the oxide layers can be silicon oxide layers, and the nitride layers can be silicon nitride layer.

The alternating dielectric stack 201 can include any suitable number of layers of the oxide layers 214 and the nitride layers 218. In some embodiments, a total number of layers of the oxide layers 214 and the nitride layers 218 in the alternating dielectric stack 201 is equal to or larger than 64. That is, a number of oxide/nitride layer pairs can be equal to or larger than 32. In some embodiments, alternating oxide/nitride stack includes more oxide layers or more nitride layers with different materials and/or thicknesses than the oxide/nitride layer pair. For example, a bottom layer and a top layer in the alternating dielectric stack 201 can be oxide layers 214.

Referring to FIG. 1B and FIG. 3, operation S100 can further include step S106 of forming an alternating dielectric

etch stop structure 205 in the periphery region 110 and forming a staircase structure 240 in the staircase and array region 120.

In some embodiments as shown in FIG. 3, a portion of the alternating dielectric stack 201 in the peripheral region 300 can be removed, such that the remaining portion of the alternating dielectric stack 201 in the peripheral region 300 forms the alternating dielectric etch stop structure 205. The process for forming the alternating dielectric etch stop structure 205 can include multiple processes, for example, anisotropic etching processes, photolithography, etc. The alternating dielectric etch stop structure 205 can include at least two dielectric layer pairs that are located at bottom of the alternating dielectric stack 201. That is, the alternating dielectric etch stop structure 205 can include two or more first dielectric layers 214 and two or more second dielectric layers 218, in the lateral direction parallel to the surface of the first substrate 101, an area of the alternating dielectric etch stop structure 205 can correspond to at least one vertical through contact to be formed in the subsequent processes.

In some embodiments, as shown in FIG. 3, multiple etch-trim processes can be performed repeatedly to form the staircase structure 240 including a set of steps at the edge the alternating dielectric stack 201. In some embodiments, each step can include one or more dielectric layer pairs. As such, each step can expose a portion of the top surface of one second dielectric layer 218. In some embodiments, the etch-trim processes can include a set of repeating etch-trim processes to form the staircase structure 240 including a set of steps at the edge the alternating dielectric stack 201.

Specifically, for forming each step, a photoresist layer (not shown) can be used as a mask to expose a portion of the top surface of the alternating dielectric stack 201. For forming the first step, a width of the exposed top surface of the alternating dielectric stack 201 can be a step width. In some embodiments, an anisotropic etching process, such as a reactive ion etching (RIE) process, or other suitable dry/wet etching process, can be performed to remove the exposed layer (e.g., the second dielectric layer 218) that is exposed through the mask (i.e., the photoresist layer). The etching process can stop on the next lower layer (e.g., the first dielectric layer 214). The pattern in the mask (i.e., the photoresist layer) is then transferred to the layer (e.g., the second dielectric layer 218) that has been etched. The exposed next lower layers (e.g., the first dielectric layers 214) can be then removed by another etching process that stops on the next lower layers (e.g., the second dielectric layer 218). As such, the first step can be created on the first two top layers of the alternating dielectric stack 201.

Next, the mask (i.e., the photoresist layer) can be reduced in size by removing a portion of the mask (also known as “trimming”) above the alternating dielectric stack 201, such as by an isotropic etching process, to expose another step width of the alternating dielectric stack 201. The method can proceed by subjecting the structure to two anisotropic etching processes, including removing exposed portions of the two exposed layers (e.g., two second dielectric layers 218), and subsequently removing exposed portions of the two exposed next lower layers (e.g., the first dielectric layers 214). As such, the first step can be lowered to the third and fourth top layers of the alternating dielectric stack 201, and a second step can be created on the first two top layers of the alternating dielectric stack 201.

In some embodiments, the successive reduction in size of the mask (i.e., the photoresist layer) and the two-step etching processes (also referred as etch-trim processes) can be repeated such that the staircase structure 240 including a set

of steps can be formed in the staircase and array region **120**, as shown in FIG. **3**. The photoresist layer can be then removed. In some embodiments, the removal process can include any suitable etching processes and cleaning processes.

In some embodiments, the alternating dielectric etch stop structure **205** can be formed simultaneously with the staircase structure **240**. For example, the mask for forming the staircase structure **240** can also initially cover the area to form the alternating dielectric etch stop structure **205** during the formation of the first two steps. As such, the alternating dielectric stack **201** in the periphery region **110** initially covered by the mask can always has two steps higher than the uncovered area of the alternating dielectric stack **201** in the periphery region **110**. Thus, after the multiple etch-trim processes to form the staircase structure **240**, the alternating dielectric etch stop structure **205** including two dielectric layer pairs as shown in FIG. **3** can also be formed.

Further, as shown in FIG. **3**, an insulating layer **250** can be formed in the periphery region **110** and the staircase and array region **120** to cover the alternating dielectric etch stop structure **205** and the staircase structure **240**. In some embodiments, a deposition process can be performed to form the insulating layer **250**, such that the insulating layer **250** can cover the exposed surfaces of the first substrate **101**, the alternating dielectric etch stop structure **205**, the staircase structure **240**, and/or the un-etched portion of the alternating dielectric stack **201**. A CMP process can be performed to planarize the top surface of the insulating layer **250**.

Referring to FIG. **1B** and FIG. **4**, operation **S100** can further include step **S108** of forming an array device **220** in the staircase and array region **120**. In some embodiments, the array device **220** can be an NAND flash memory device in which memory cells are provided in the form of an array of NAND strings **230** extending vertically above first substrate **101**. The array device **220** can include a plurality of NAND strings **230** that extend through a plurality of conductor layer **212** and dielectric layer **214** pairs. The plurality of conductor/dielectric layer pairs are also referred to herein as an “alternating conductor/dielectric stack” **210**. The alternating conductor/dielectric stack **210** can be formed by replacing the multiple second dielectric layers **218** in the alternating dielectric stack **201** and the staircase structure **240** with multiple conductor layers **212** to form the word lines. Thus, the dielectric layer pairs can be converted into conductor/dielectric layer pairs. The conductor layers **212** can include conductor materials including, but not limited to, tungsten (W), cobalt (Co), copper (Cu), aluminum (Al), polycrystalline silicon (polysilicon), doped silicon, silicides, or any combination thereof.

As shown in FIG. **4**, after the replacement, each level of staircase structure **240** can include one or more conductor/dielectric layer pairs, each including at least one conductor layer **212** and at least one first dielectric layer **214**. The top layer in each level of staircase structure **240** can be the conductor layer **212** for interconnection in the vertical direction. In some embodiments, each two adjacent levels of staircase structure **240** are offset by a nominally same distance in the vertical direction and a nominally same distance in the lateral direction. For each two adjacent levels of staircase structure **240**, the first level (and conductor layer and dielectric layer therein) that is closer to first substrate **101** can extend laterally further than the second level (and conductor layer and dielectric layer therein), thereby forming a “landing area” on the first level for interconnection in the vertical direction.

As shown in FIG. **3**, each NAND string **230** can include a channel structure extending through alternating conductor/dielectric stack. The channel structure can include a channel hole filled with semiconductor materials (e.g., as a semiconductor channel) and dielectric materials (e.g., as a memory film). In some embodiments, the semiconductor channel includes silicon, such as amorphous silicon, polysilicon, or single crystalline silicon. In some embodiments, the memory film is a composite layer including a tunneling layer, a storage layer (also known as “charge trap/storage layer”), and a blocking layer. Each NAND string **230** can have a cylinder shape (e.g., a pillar shape). The semiconductor channel, the tunneling layer, the storage layer, and the blocking layer are arranged along a direction from the center toward the outer surface of the pillar in this order, according to some embodiments. The tunneling layer can include silicon oxide, silicon nitride, or any combination thereof. The storage layer can include silicon nitride, silicon oxynitride, silicon, or any combination thereof. The blocking layer can include silicon oxide, silicon nitride, high dielectric constant (high-k) dielectrics, or any combination thereof.

In some embodiments, NAND strings **230** include a plurality of control gates (each being part of a word line) for NAND strings **230**. Conductor layer **212** in alternating conductor/dielectric stack **210** can function as a control gate for memory cells of NAND string **230**. Conductor layer **212** can include multiple control gates for multiple NAND strings **230** and can extend laterally as a word line ending in staircase structure **240**.

As shown in FIG. **1B** and FIG. **5**, operation **S100** can further include step **S110** of forming a plurality of vertical through contacts **310** in the insulating layer **250**. As shown in FIG. **4**, the plurality of vertical through contacts **310** can be formed in the insulating layer **250**, and can include at least one first vertical contact **312** in contact with the alternating dielectric etch stop structure **205**, at least one second vertical contact **314** in contact with the array well structure **114**, and multiple word line contacts **316** in contact with the word lines in staircase structure **240**. Each vertical through contact **310** can include an opening (e.g., a via hole or a trench) filled with conductor materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. Some or all of the plurality of vertical through contacts **310** can be formed simultaneously in a single contact forming process as described below in detail.

In some embodiments, an end (e.g., the upper end) of each vertical through contact **310** is flush with one another, e.g., on the top surface of the insulating layer **250** in which the vertical through contacts **310** are formed. Another end (e.g., the lower end) of each vertical through contact **310** can be in contact with the respective array wafer structure. For example, the lower end of the first vertical contact **312** can be in contact with the alternating dielectric etch stop structure **205**, the lower end of the second vertical contact **314** can be in contact with the array well structure **114**, and the lower end of each word line contacts **316** can be in contact with corresponding conductor layer **206** (word line) in one level of staircase structure **240**.

It is understood that, a contact forming process for forming the plurality of vertical through contacts **310** can include multiple processes, for example, photolithography, etching, thin film deposition, and CMF. For example, vertical through contacts **310** can be formed through insulating layer **250** by first deep etching vertical openings (e.g., by wet etching and/or dry etching), followed by filling the openings with conductor materials using ALD, CVD, PVD, any other suitable processes, or any combination thereof. The conduc-

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tor materials used for filling the vertical through contacts **310** can include, but are not limited to, W, Co, Cu, Al, polysilicon, silicides, or any combination thereof. In some embodiments, other conductor materials are also used to fill the openings to function as a barrier layer, an adhesion layer, and/or a seed layer.

In some embodiments, the plurality of vertical through contacts **310** including the first vertical contacts **312**, the second vertical contacts **314**, and the word line contacts **316**, can be simultaneously formed in the same contact forming process. In some embodiments, each process in the contact forming process needs to be performed only once for all of the first vertical contacts **312**, the second vertical contacts **314**, and the word line contacts **316**. For example, a single lithography process can be performed to pattern the masks for all the openings of the first vertical contacts **312**, the second vertical contacts **314**, and the word line contacts **316**; a single etching process can be performed to etch all the openings of the first vertical contacts **312**, the second vertical contacts **314**, and the word line contacts **316**; a single deposition process can be performed to fill all the openings of the first vertical contacts **312**, the second vertical contacts **314**, and the word line contacts **316** with the same conductor materials.

As shown in FIG. 1B and FIG. 5, operation **S100** can further include step **S112** of forming at least one contact layer **320** including a plurality of interconnect contacts **323**. The contact layer **320** can include a dielectric layer **325** and interconnect contacts **323**, and can be formed above the insulating layer **250**. The upper end of each interconnect contact **323** can be flush with one another at the top surface of dielectric layer **325**, and the lower end of each interconnect contact **323** can be flush with one another at the bottom surface of the dielectric layer **325**, and can be in contact with the upper end of a corresponding vertical through contact **310**, for example, the first vertical contact **312**, the second vertical contact **314**, or the word line contact **316**.

The dielectric layer **325** can be formed by one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. Dielectric layer **325** can include dielectric materials, including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. The interconnect contacts **323** can be formed through dielectric layer **325** by first etching vertical openings (e.g., by wet etching and/or dry etching), followed by filling the openings with conductor materials using ALD, CVD, PVD, any other suitable processes, or any combination thereof. The conductor materials used to fill the interconnect contacts **323** can include, but are not limited to, W, Co, Cu, Al, polysilicon, silicides, or any combination thereof. In some embodiments, other conductor materials are used to fill the openings to function as a barrier layer, an adhesion layer, and/or a seed layer.

In some embodiments, the at least one contact layer **320** can include multiple layers, and each interconnect contact **323** can include multiple sub-contacts formed in the multiple layers. For example, as shown in FIG. 4, the multiple sub-contacts can include one or more contacts, single-layer/multi-layer vias, conductive lines, plugs, pads, and/or any other suitable conductive structures that are made by conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof, and can be formed in multiple contact forming processes. For example, fabrication processes to form the multiple sub-contacts can include forming one or more conductive layers and one or more contact layer in the dielectric layer **325**. The conductive layers and the conductor contact layers can be

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formed by any suitable known back-end-of-line (BEOL) methods. In some embodiments, all interconnect contacts **323** in contact layer **320** can be simultaneously formed in the same contact forming processes.

As shown in FIG. 1B and FIG. 5, operation **S100** can further include step **S114** of forming an array joint layer **330** on the at least one contact layer **320**. The array joint layer **330** can be a back-end-of-line (BEOL) interconnection layer including one or more joint structures **338** embedded in a dielectric layer **336**. The joint structures **338** can include, but not limited to, contacts, single-layer/multi-layer vias, conductive lines, plugs, pads, and/or any other suitable conductive structures that are made by conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The dielectric layer **336** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. One or more portions of the joint structures **338** can be exposed on the top surface of the array joint layer **330** on the array wafer **100**.

In some embodiments, fabrication processes to form the array joint layer **330** include forming a dielectric layer **336**, followed by forming a plurality of joint structures **338**. One or more of the joint structures **338** can be in contact with the interconnect contacts **323** respectively. The dielectric layer **336** can include one or more layers of dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. The joint structures **338** can include, but not limited to, contacts, single-layer/multi-layer vias, conductive lines, plugs, pads, and/or any other suitable conductive structures that are made by conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof.

In some embodiments, fabrication processes to form the joint structures **338** include forming openings in the dielectric layer **336**, followed by filling the openings with conductive materials. The openings in the dielectric layer **336** can be filled with conductive materials by ALD, CVD, PVD, any other suitable processes, or any combination thereof. In some embodiments, fabrication processes to form the joint structures **338** further include forming one or more conductive layers and one or more contact layer in the dielectric layer. The conductive layers and the conductor contact layers can be formed by any suitable known BEOL methods.

Referring back to FIG. 1, the method processed to operation **S200**, in which an CMOS wafer can be formed. As shown in FIG. 6, the CMOS wafer **400** can include a second substrate **401**, a peripheral circuit layer **410** on the second substrate **401**, at least one contact layer **420** on the peripheral circuit layer **410**, and a CMOS joint layer **430** on the at least one contact layer **320**.

As shown in FIG. 1C and FIG. 6, operation **S200** can include step **S202** of forming a second substrate **401**, and forming a peripheral circuit layer **410** on the second substrate **401**. In some embodiments, the second substrate **401** can include any suitable semiconductor material that can include silicon (e.g., monocrystalline silicon, polycrystalline silicon), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon on insulator (SOI), germanium on insulator (GOI), or any suitable combination thereof. Second substrate **401** can be a single layer substrate or a multi-layer substrate, for example, a monocrystalline single-layer substrate, a polycrystalline silicon (polysilicon) single-layer substrate, a polysilicon and metal multi-layer substrate, etc. In some embodiments, second substrate **401** is a thinned substrate (e.g., a semiconductive layer), which was

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thinned by grinding, wet/dry etching, chemical mechanical polishing (CMP), or any combination thereof.

The peripheral circuit layer **410** formed on the second substrate **401** can include one or more peripheral circuits including any suitable digital, analog, and/or mixed-signal peripheral circuits used for facilitating the operation of the 3D memory device. For example, the one or more peripheral circuits can include one or more of a page buffer, a decoder (e.g., a row decoder and a column decoder), a latch, a sense amplifier, a driver, a charge pump, a current or voltage reference, or any active or passive components of the circuits (e.g., transistors, diodes, resistors, or capacitors). In some embodiments as shown in FIG. 6, the one or more peripheral circuits can include multiple CMOS devices **412** formed using complementary metal-oxide-semiconductor (CMOS) technology.

As shown in FIG. 1C and FIG. 6, operation **S200** can further include step **S204** of forming at least one contact layer **420** including a plurality of interconnect contacts **423**. The contact layer **420** can include a dielectric layer **425** and interconnect contacts **423**, and can be formed above the peripheral circuit layer **410**. The dielectric layer **425** can be formed by one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. Dielectric layer **425** can include dielectric materials, including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. The interconnect contacts **423** can be formed through dielectric layer **425** by first etching vertical openings (e.g., by wet etching and/or dry etching), followed by filling the openings with conductor materials using ALD, CVD, PVD, any other suitable processes, or any combination thereof. The conductor materials used to fill the interconnect contacts **423** can include, but are not limited to, W, Co, Cu, Al, polysilicon, silicides, or any combination thereof. In some embodiments, other conductor materials are used to fill the openings to function as a barrier layer, an adhesion layer, and/or a seed layer.

In some embodiments, the at least one contact layer **420** can include multiple layers, and each interconnect contact **423** can include multiple sub-contacts formed in the multiple layers. For example, as shown in FIG. 6, the multiple sub-contacts can include one or more contacts, single-layer/multi-layer vias, conductive lines, plugs, pads, and/or any other suitable conductive structures that are made by conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof, and can be formed in multiple contact forming processes. For example, fabrication processes to form the multiple sub-contacts can include forming one or more conductive layers and one or more contact layer in the dielectric layer **425**. The conductive layers and the conductor contact layers can be formed by any suitable known front-end-of-line (FEOL) methods. In some embodiments, all interconnect contacts **423** in contact layer **420** can be simultaneously formed in the same contact forming processes.

As shown in FIG. 1C and FIG. 6, operation **S200** can further include step **S206** of forming a CMOS joint layer **430** on the at least one contact layer **420**. The CMOS joint layer **430** can be a FOEL interconnection layer including one or more joint structures **438** embedded in a dielectric layer **436**. The joint structures **438** can include, but not limited to, contacts, single-layer/multi-layer vias, conductive lines, plugs, pads, and/or any other suitable conductive structures that are made by conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The dielectric layer **436** can include

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dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. One or more portions of the joint structures **438** can be exposed on the top surface of the CMOS joint layer **430** on the CMOS wafer **400**.

In some embodiments, fabrication processes to form the CMOS joint layer **430** include forming a dielectric layer **436**, followed by forming a plurality of joint structures **438**. One or more of the joint structures **438** can be in contact with the interconnect contacts **423** respectively. The dielectric layer **436** can include one or more layers of dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. The joint structures **438** can include, but not limited to, contacts, single-layer/multi-layer vias, conductive lines, plugs, pads, and/or any other suitable conductive structures that are made by conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof.

In some embodiments, fabrication processes to form the joint structures **438** include forming openings in the dielectric layer **436**, followed by filling the openings with conductive materials. The openings in the dielectric layer **436** can be filled with conductive materials by ALD, CVD, PVD, any other suitable processes, or any combination thereof. In some embodiments, fabrication processes to form the joint structures **438** further include forming one or more conductive layers and one or more contact layer in the dielectric layer. The conductive layers and the conductor contact layers can be formed by any suitable known FEOL methods.

Referring back to FIG. 1A, the method processed to operation **S300**, in which the array wafer and the CMOS wafer can be bonded. As shown in FIG. 1D and FIG. 7, operation **S300** can include step **S302** of flipping over the array wafer **100** to face down towards the CMOS wafer **400** for hybrid bonding, and step **S304** of bonding the array joint layer **330** of the array wafer **100** to the CMOS joint layer **430** of the CMOS wafer **400**.

The bonding interface is between the array joint layer **330** and the CMOS joint layer **430**. Therefore, the bonding interface includes both the interface between two dielectric layers (e.g., between a silicon nitride layer and a silicon oxide layer) and the interface between two conductive layers (e.g., between two metal layers). In some embodiments, one or more joint structures **338** in the array wafer **100** and one or more joint structures **438** in the CMOS wafer **400** can be contacted with each other at bonding interface for electrical connection.

In some embodiments, the hybrid bonding between the array wafer **100** and the CMOS wafer **400** can include any suitable bonding processes or combinations thereof. For example, the bonding interface can be formed by chemical bonds between the dielectric layers and/or the conductive layers on both sides of the bonding interface. As another example, the bonding interface can be formed by physical interaction (e.g., inter-diffusion) between the dielectric layers and/or the conductive layers on both sides of the bonding interface. In some embodiments, the bonding interface can be formed after a plasma treatment or a thermal treatment of the surfaces from both sides of the bonding interface prior to the bonding process.

Referring back to FIG. 1A, the method processed to operation **S400**, in which at least one through substrate contact can be formed to penetrate the first substrate and the alternating dielectric etch stop structure to connect with the at least one first vertical contact.

As shown in FIG. 1F, operation **S400** can include step **S402** of thinning the first substrate. In some embodiments,

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the first substrate **101** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

As shown in FIG. 1E and FIG. 8, operation **S400** can include step **S404** of forming at least one through substrate opening **610** that penetrates the first substrate **101**. It is understood that, the at least one through substrate opening **610** can be formed by any suitable photolithography process. For example, a hard mask can be formed on the first substrate **101**, and a deep plasma etching (e.g., deep reactive-ion etching process) can be performed to form the at least one through substrate opening **610** penetrating the entire first substrate **101** and stopped at the alternating dielectric etch stop structure **205**.

It is noted that, during the plasma etching process, a high-energy stream of glow discharge (plasma) of an appropriate gas mixture being shot (in pulses) at a sample. The plasma source (also referred as "etch species") can be either charged (ions) or neutral (atoms and radicals). During the plasma etching process, the plasma generates volatile etch products from the chemical reactions between the first substrate **101** and the reactive species generated by the plasma. It is also understood that, during the plasma etching process, the high-energy stream of plasma is blocked by the alternating dielectric etch stop structure **205** and the array well structure **114**, and thus cannot go through the at least one first vertical contact **312** and/or the at least one second vertical contact **314** to affect the CMOS devices in the CMOS wafer **400**. Therefore, the potential plasma-induced damage (PID) to the CMOS devices in the CMOS wafer **400** can be avoided.

As shown in FIG. 1E and FIG. 9, operation **S400** can further include step **S406** of forming an isolating layer **730** covering the first substrate **101** and filling the at least one through substrate opening **610**. The isolating layer **730** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof.

As shown in FIG. 1E and FIG. 9, operation **S400** can further include step **S408** of forming at least one through substrate contact **740** (also referred as "through silicon contact (TSC)" in some embodiments) that penetrates the isolating layer **730** and the alternating dielectric etch stop structure **205**, and in contact with the at least one first vertical contact **312**. It is understood that, a contact process for forming the at least one through substrate contact **740** can include multiple processes, for example, photolithography, etching, thin film deposition, and CMP. In some embodiments, at least one vertical through opening can be formed through corresponding through substrate opening **610** by a wet etching and/or dry etching to penetrate the dielectric materials of the isolating layer **730** and the alternating dielectric etch stop structure **205**. As such, the end surface of the at least one first vertical contact **312** that are landed at the alternating dielectric etch stop structure **205** can be exposed by the at least one vertical through opening. A followed deposition process can form the at least one through substrate contact **740** by filling the at least one vertical through opening with conductor materials using ALD, CVD, PVD, any other suitable processes, or any combination thereof. The conductor materials used for filling the at least one through substrate contact **740** can include, but are not limited to, W, Co, Cu, Al, polysilicon, silicides, or any combination thereof. In some embodiments, other conductor materials are also used to fill the openings to function as a barrier layer, an adhesion layer, and/or a seed layer.

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As shown in FIG. 1F, and FIG. 10, operation **S400** can further include step **S410** of forming at least one array pad **820** in contact with the least one through substrate contact **740**. It is understood that, a contact process for forming the at least one array pad **820** can include multiple processes, for example, photolithography, etching, thin film deposition, and CMP. In some embodiments, a hard mask layer can be formed on the isolating layer **730**, and at least one vertical through opening can be formed in the isolating layer **730** by a wet etching and/or dry etching by using the hard mask layer. As such, the end surface of the least one through substrate contact **740** can be exposed by the at least one vertical through opening. A followed deposition process can form at least one array pad **820** by filling the at least one vertical through opening with conductor materials using ALD, CVD, PVD, any other suitable processes, or any combination thereof. The conductor materials used for filling the at least one array pad **820** can include, but are not limited to, W, Co, Cu, Al, polysilicon, silicides, or any combination thereof. In some embodiments, other conductor materials are also used to fill the openings to function as a barrier layer, an adhesion layer, and/or a seed layer. A protecting layer **810** can be formed on the isolating layer **730**, and at least one pad opening **830** can be formed to expose the at least one array pad **820**.

By using the at least one array pad **820** on the back side of the first substrate, rather than the surrounding regions of the first substrate, the periphery circuits of the 3D memory device can be connected out through the back side of the 3D memory device. Thus, the size of the 3D memory device can be reduced and the integration degree of the 3D memory device can be increased. Further, the risk of plasma-induced damage (PID) to the CMOS devices can be eliminated by using the alternating dielectric etch stop structure and the array well structure to block plasma during forming the vertical through contacts, thereby increasing the CMOS device.

Accordingly, a 3D memory device and a fabricating method thereof are provided. In some embodiments, the method can include forming an array wafer including a periphery region and a staircase and array region, comprising: forming an alternating dielectric etch stop structure on a first substrate in the periphery region, forming an array device on the first substrate in the staircase and array region, and forming at least one first vertical through contact in the periphery region and in contact with the alternating dielectric etch stop structure. The method can further include: forming a CMOS wafer; bonding the array wafer and the CMOS wafer; and forming at least one through substrate contact penetrating the first substrate and the alternating dielectric etch stop structure, and in contact with the at least one first vertical through contact.

In some embodiments, forming the array wafer further comprises: forming an array well structure in the first substrate in the periphery region; and forming at least one second vertical through contact in contact with the array well structure.

In some embodiments, forming the array wafer further comprises: forming an alternating dielectric stack on the first substrate; and removing a portion of the alternating dielectric stack to simultaneously form the alternating dielectric etch stop structure in the periphery region and a staircase structure on at least one lateral side of the alternating dielectric stack in the staircase and array region.

In some embodiments, forming the array device comprises: converting the alternating dielectric stack in the staircase and array region to an alternating conductor/di-

electric stack; and forming a plurality of NAND strings vertically penetrating the alternating conductor/dielectric stack.

In some embodiments, forming the array wafer further comprises: forming an insulating layer covering the alternating dielectric etch stop structure, the array well structure, and the array device; and forming at least one word line contact in the staircase and array region and in contact with a word line in the staircase structure; wherein the at least one first vertical through contact, the at least one second vertical through contact, and the at least one word line contact are simultaneously formed in the insulating layer by a same contact forming process.

In some embodiments, forming the array wafer further comprises: forming at least one first contact layer including a plurality of first interconnect contacts on the insulating layer; and forming an array joint layer on the at least one first contact layer.

In some embodiments, forming the CMOS wafer comprises: forming a peripheral circuit layer on a second substrate; forming at least one second contact layer including a plurality of second interconnect contacts on the peripheral circuit layer; and forming a CMOS joint layer on the at least one second contact layer.

In some embodiments, bonding the array wafer to the CMOS wafer comprises: flipping over the array wafer to face down towards the CMOS wafer; and bonding the array joint layer of the array wafer to the CMOS joint layer of the CMOS wafer, such that the at least one first vertical through contact is electrically connected to the peripheral circuit layer through at least one first interconnect contact and at least one second interconnect contact.

In some embodiments, forming the at least one through substrate contact comprises: forming at least one through substrate opening penetrating the first substrate; forming an isolating layer covering the first substrate and filling the at least one through substrate opening; forming at least one vertical through opening that penetrates the isolating layer, the at least one through substrate opening, and the alternating dielectric etch stop structure, and exposes at least a portion of the at least one first vertical through contact; and forming the at least one through substrate contact in the at least one vertical through opening, such that the at least one through substrate contact is in contact with the at least one first vertical contact.

In some embodiments, forming at least one through substrate opening comprises: using a deep plasma etching to form the at least one through substrate opening in the first substrate; wherein a high-energy stream of plasma during the deep plasma etching is blocked by the alternating dielectric etch stop structure and the array well structure.

In some embodiments, forming the at least one through substrate contact further comprises: forming at least one array pad in contact with the least one through substrate contact; and forming at least one pad opening to expose the at least one array pad.

The disclosed three-dimensional (3D) memory device comprises an array wafer including a periphery region and a staircase and array region, comprising: a first substrate, an alternating dielectric etch stop structure on the first substrate in the periphery region, an array device on the first substrate in the staircase and array region, at least one first vertical through contact in the periphery region, and at least one through substrate contact penetrating the first substrate and the alternating dielectric etch stop structure, and in contact with the at least one first vertical through contact. The 3D memory device further comprises a CMOS wafer bonded on

the array wafer, comprising a peripheral circuit layer electrically connected to the at least one through substrate contact through the at least one first vertical through contact.

In some embodiments, the array wafer further comprises: an array well structure in the first substrate in the periphery region; and at least one second vertical through contact in contact with the array well structure.

In some embodiments, the alternating dielectric etch stop structure comprises at least two dielectric layer pairs each including a first dielectric layer and a second dielectric layer different from the first dielectric layer.

In some embodiments, the array device comprises: an alternating conductor/dielectric stack on the first substrate; a plurality of NAND strings vertically penetrating the alternating conductor/dielectric stack; and a staircase structure on at least one lateral side of the alternating conductor/dielectric stack.

In some embodiments, the array wafer further comprises: an insulating layer covering the alternating dielectric etch stop structure, the array well structure, and the array device; at least one word line contact in the staircase and array region and in contact with a word line in the staircase structure; wherein the at least one first vertical through contact, the at least one second vertical through contact, and the at least one word line contact penetrate the insulating layer.

In some embodiments, the array wafer further comprises: at least one first contact layer including a plurality of first interconnect contacts covering the insulating layer; and an array joint layer between the at least one first contact layer and the CMOS wafer.

In some embodiments, the CMOS wafer comprises: a peripheral circuit layer on a second substrate; at least one second contact layer including a plurality of second interconnect contacts on the peripheral circuit layer; and a CMOS joint layer between the at least one second contact layer and the array joint layer.

In some embodiments, the array wafer further comprises: an isolating layer covering the first substrate; wherein the at least one through substrate contact penetrates the isolating layer and the alternating dielectric etch stop structure, and in contact with the at least one first vertical contact.

In some embodiments, the array wafer further comprises: at least one array pad in contact with the least one through substrate contact; wherein the at least one array pad is electrically connected to the peripheral circuit layer of the CMOS wafer through the at least one first vertical through contact, the at least one first interconnect contact, and the at least one second interconnect contact.

Other aspects of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.

The foregoing description of the specific embodiments will so fully reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

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Embodiments of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

What is claimed is:

1. A method for forming a three-dimensional (3D) memory device, comprising:

forming an etch stop structure in a first wafer, the etch stop structure comprising at least two dielectric layer pairs, each dielectric layer pair including a first dielectric layer and a second dielectric layer;

forming a first through contact disposed on and in direct contact with a top dielectric surface of the etch stop structure;

bonding the first wafer to a second wafer to electrically connect the first through contact to a CMOS device of the second wafer; and

forming a through substrate contact penetrating a first substrate of the first wafer and through the etch stop structure, and in electrically contact with the CMOS device through the first through contact.

2. The method of claim 1, wherein forming the etch stop structure comprises forming coplanar side surfaces of an alternating dielectric stack structure that includes the at least two dielectric layer pairs.

3. The method of claim 1, further comprising: forming an array well structure in the first substrate; and forming a second through contact in contact with the array well structure.

4. The method of claim 3, further comprising: forming an alternating dielectric stack including the at least two dielectric layer pairs on the first substrate; and removing a portion of the alternating dielectric stack to simultaneously form the etch stop structure and a staircase structure.

5. The method of claim 4, further comprising: converting the staircase structure to an alternating conductor/dielectric stack; and

forming a plurality of NAND strings vertically penetrating the alternating conductor/dielectric stack.

6. The method of claim 5, further comprising: forming an insulating layer covering the etch stop structure, the array well structure, and the alternating conductor/dielectric stack; and

forming a word line contact and in contact with a word line in the staircase structure;

wherein the first through contact, the second through contact, and the word line contact are simultaneously formed in the insulating layer by a same contact forming process.

7. The method of claim 6, further comprising: forming a first contact layer including a plurality of first interconnect contacts on the insulating layer; and forming an array joint layer on the first contact layer.

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8. The method of claim 7, further comprising:

forming a peripheral circuit layer on a second substrate of the second wafer;

forming a second contact layer including a plurality of second interconnect contacts on the peripheral circuit layer; and

forming a CMOS joint layer on the second contact layer.

9. The method of claim 8, wherein bonding the first wafer to the second wafer comprises:

bonding the array joint layer of the first wafer to the CMOS joint layer of the second wafer, such that the first through contact is electrically connected to the peripheral circuit layer through at least one first interconnect contact and at least one second interconnect contact.

10. The method of claim 9, wherein forming the through substrate contact comprises:

forming a through substrate opening penetrating the first substrate;

forming an isolating layer covering the first substrate and filling the through substrate opening;

forming a vertical through opening that penetrates the isolating layer, the through substrate opening, and the etch stop structure, and exposes at least a portion of the first through contact; and

forming the through substrate contact in the vertical through opening, such that the through substrate contact is in contact with the first through contact.

11. The method of claim 10, wherein forming the through substrate opening comprises:

using a deep plasma etching to form the through substrate opening in the first substrate;

wherein a high-energy stream of plasma during the deep plasma etching is blocked by the etch stop structure and the array well structure.

12. The method of claim 10, wherein forming the through substrate contact further comprises:

forming an array pad in contact with the through substrate contact; and

forming a pad opening to expose the array pad.

13. A three-dimensional (3D) memory device, comprising:

a first wafer comprising:

a first substrate;

an etch stop structure on the first substrate, the etch stop structure comprising at least two dielectric layer pairs, each dielectric layer pair including a first dielectric layer and a second dielectric layer, the first dielectric layer on the first substrate, wherein widths of the first and second dielectric layers are substantially the same;

a first through contact on the etch stop structure;

a through substrate contact penetrating the first substrate and through the first and second dielectric layers, wherein the through substrate contact is in contact with the first through contact; and

a second wafer bonded on the first wafer, comprising a peripheral circuit layer electrically connected to the through substrate contact via the first through contact.

14. The 3D memory device of claim 13, wherein the first wafer further comprises:

an array well structure in the first substrate; and

a second through contact in contact with the array well structure.

15. The 3D memory device of claim 13, wherein the etch stop structure further comprises:

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a third dielectric layer on the first dielectric layer and under the second dielectric layer.

16. The 3D memory device of claim **14**, wherein the first wafer further comprises:

an alternating conductor/dielectric stack on the first substrate; 5

a plurality of NAND strings vertically penetrating the alternating conductor/dielectric stack; and

a staircase structure on a lateral side of the alternating conductor/dielectric stack. 10

17. The 3D memory device of claim **16**, wherein the first wafer further comprises:

a word line contact in contact with a word line in the staircase structure;

wherein the first through contact, the second through contact, and the word line contact penetrate an insulating layer covering the etch stop layer. 15

18. The 3D memory device of claim **16**, wherein the first wafer further comprises:

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a first contact layer covering an insulating layer covering the etch stop layer and including a first interconnect contact; and

an array joint layer between the first contact layer and the second wafer.

19. The 3D memory device of claim **18**, wherein the second wafer comprises:

a second contact layer including a second interconnect contact on the peripheral circuit layer; and

a CMOS joint layer between the second contact layer and the array joint layer.

20. The 3D memory device of claim **19**, wherein the first wafer further comprises:

a pad in contact with the through substrate contact;

wherein the pad is electrically connected to the peripheral circuit layer of the second wafer through the first through contact, the first interconnect contact, and the second interconnect contact.

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